

Lakshya JEE 2.0 (2025)

Mathematics

DPP: 5

Indefinite Integration

Q1 If $\int x \sin x dx = -x \cos x + A$, then $A =$

- (A) $\sin x + \text{constant}$
 (B) $\cos x + \text{constant}$
 (C) Constant
 (D) None of these

Q2 $\int (\log x)^2 dx$ equals

- (A) $x(\log x)^2 - 2x \log x - 2x + C$
 (B) $x(\log x)^2 - 2x \log x - x + C$
 (C) $x(\log x)^2 - 2x \log x + 2x + C$
 (D) $x(\log x)^2 - 2x \log x + x + C$

Q3 $\int \frac{\log x}{x^2} dx$

- (A) $\frac{1}{2}(1 + \log x) + C$ (B) $x(1 + \log x) + C$
 (C) $-x(1 + \log x) + C$ (D) $\frac{-1}{x}(1 + \log x) + C$

Q4 If $\int x^2 e^{-2x} dx = e^{-2x} (ax^2 + bx + c) + d$, then the value of $|a/bc|$ is

- (A) 1 (B) 2
 (C) 3 (D) 4

Q5 If $\int (\sin 3\theta + \sin \theta) e^{\sin \theta} \cos \theta d\theta$

$= (A \sin^3 \theta + B \cos^2 \theta + C \sin \theta + D \cos \theta + E) e^{\sin \theta} + F$,
 then

- (A) $B = 12$ (B) $D = 0$
 (C) $B = -12$ (D) None of these

Q6 $\int \sqrt{\cot \theta} d\theta$ Q7 $\int e^{3x} \sin 2x dx$ equals

- (A) $\frac{e^{3x}}{13} (3 \cos 2x + 2 \sin 2x) + C$
 (B) $\frac{e^{3x}}{13} (3 \sin 2x - 2 \cos 2x) + C$
 (C) $\frac{e^{3x}}{13} (-3 \sin 2x - 2 \cos 2x) + C$
 (D) $\frac{e^{3x}}{13} (3 \sin 2x + 2 \cos 2x) + C$

Q8 $\int e^x (\tan x + \log \sec x) dx$

Q9 If

$$\int \sin^{-1} \left(\sqrt{\frac{x}{1+x}} \right) dx = A(x) \cdot \tan^{-1} \sqrt{x}, \text{ where } + B(x) + c$$

c is a constant of integration, then the ordered pair $(A(x), B(x))$ can be:

- (A) $(x+1, -\sqrt{x})$
 (B) $(x+1, \sqrt{x})$
 (C) $(x-1, -\sqrt{x})$
 (D) $(x-1, \sqrt{x})$

Q10 $\int x \cdot 2^x dx$ Q11 $\int x^3 \sin(x^2) dx$ Q12 $\int \frac{e^x}{x} (I + x \cdot \ln x) dx$ Q13 $\int [\sin(\ln x) + \cos(\ln x)] dx$ Q14 $\int \frac{x^2 e^x}{(x+2)^2} dx$ Q15 Evaluate: $\int x e^{x^2} (\sin x^2 + \cos x^2) dx$ Q16 The value of $\int e^x \left(\frac{1-x}{1+x^2} \right)^2 dx$ is

- (A) $e^x \left(\frac{1-x}{1+x^2} \right) + C$
 (B) $e^x \left(\frac{1+x}{1+x^2} \right) + C$
 (C) $\frac{e^x}{1+x^2} + C$
 (D) $e^x (1-x) + C$

Q17 If $\int e^x \left(\frac{1-\sin x}{1-\cos x} \right) dx = f(x) + \text{constant}$, then

$f(x)$ is equal to

- (A) $e^x \cot\left(\frac{x}{2}\right) + C$
 (B) $e^{-x} \cot\left(\frac{x}{2}\right) + C$
 (C) $-e^x \cot\left(\frac{x}{2}\right) + C$
 (D) $-e^{-x} \cot\left(\frac{x}{2}\right) + C$



Q18 $\int e^x \left(\log x + \frac{1}{x^2} \right) dx$

Q19 $\int \left(1 + x - \frac{1}{x} \right) e^{x+\frac{1}{x}} dx$ is equal to

- (A) $(x-1)e^{x+\frac{1}{x}} + C$
 (B) $xe^{x+\frac{1}{x}} + C$
 (C) $(x+1)e^{x+\frac{1}{x}} + C$
 (D) $-xe^{x+\frac{1}{x}} + C$

Q20 $\int e^x \left[\frac{\sin x + \cos x}{1 - \sin^2 x} \right] dx$ is equal to

- (A) $e^x \cdot \operatorname{cosec} x + C$
 (B) $e^x \cdot \cot x + C$
 (C) $e^x \cdot \sec x + C$
 (D) $e^x \cdot \tan x + C$

Q21 $\int \frac{\sqrt{1-\sin x}}{1+\cos x} e^{-x/2} dx, \quad x \in (0, \frac{\pi}{2})$

- (A) $e^{-x/2} \sin \frac{x}{2} + C$
 (B) $-e^{-x/2} \sec \frac{x}{2} + C$
 (C) $-e^{-x/2} \sin \frac{x}{2} + C$
 (D) $-e^{-x/2} \cos \frac{x}{2} + C$

Q22 The value of integral

$\int e^x \left[\left(\frac{2 \tan x}{1 + \tan x} \right) + \cot^2 \left(x + \frac{\pi}{4} \right) \right] dx$ is equal to:

- (A) $e^x \tan \left(\frac{\pi}{4} - x \right) + C$
 (B) $e^x \tan \left(x - \frac{\pi}{4} \right) + C$
 (C) $e^x \tan \left(\frac{3\pi}{4} - x \right) + C$
 (D) $e^x \tan \left(x - \frac{3\pi}{4} \right) + C$

Q23 Evaluate $\int \frac{x^2 dx}{(x \sin x + \cos x)^2}$

- (A) $\frac{-x \sec x}{(x \sin x + \cos x)} + \tan x + C$
 (B) $\frac{-x \cos x}{(x \sin x + \cos x)} + \tan x + C$
 (C) $\frac{x \sec x}{(x \sin x + \cos x)} + \tan x + C$
 (D) None of these

Q24 $\int 2x^3 \cdot e^{x^2} dx.$

- (A) $x^2 \cdot e^{x^2} + e^{x^2} + c$
 (B) $-x^2 \cdot e^{x^2} - e^{x^2} + c$
 (C) $-x^2 \cdot e^{x^2} + e^{x^2} + c$
 (D) $x^2 \cdot e^{x^2} - e^{x^2} + c$

Q25 $\int \frac{\sin^{-1} x dx}{(1-x^2)^{3/2}}$

- (A)

$-\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} + \log[\sqrt{1-x^2}] + c$

(B) $\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} + \log[\sqrt{1-x^2}] + c$

(C) $\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} - \log[\sqrt{1-x^2}] + c$

(D) $-\frac{x \cdot \sin^{-1} x}{\sqrt{1-x^2}} - \log[\sqrt{1-x^2}] + c$



Answer Key

Q1 (A)

Q2 (C)

Q3 (C)

Q4 (D)

Q5 (B, C)

$$\text{Q6} \quad \frac{-1}{2\sqrt{2}} \log \left(\frac{\sqrt{\cot\theta} + \frac{1}{\sqrt{\cot\theta}} - \sqrt{2}}{\sqrt{\cot\theta} + \frac{1}{\sqrt{\cot\theta}} + \sqrt{2}} \right) + c$$

Q7 (B)

$$\text{Q8} \quad e^x \log \sec x + c$$

Q9 (A)

$$\text{Q10} \quad \frac{x \cdot 2^x}{\log 2} - \frac{2^x}{(\log 2)^2} + c$$

$$\text{Q11} \quad \frac{1}{2} [-x^2 \cos x^2 + \sin x^2] + c$$

Q12 (A)

$$\text{Q13} \quad x \sin(\log x) + c$$

$$\text{Q14} \quad e^x \left(\frac{x-2}{x+2} \right) + c$$

$$\text{Q15} \quad = \frac{1}{2} e^{x^2 \sin x^2} + c$$

Q16 (C)

Q17 (C)

$$\text{Q18} \quad e^x \left(\log x - \frac{1}{x} \right) + c$$

Q19 (B)

Q20 (C)

Q21 (B)

Q22 (B)

Q23 (A)

Q24 (D)

Q25 (B)



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